Learning Control

Part III: Flexible-Task ILC

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Introduction

Iterative Learning Control (ILC)

- Performance improvement for repeated tasks

Key assumptions

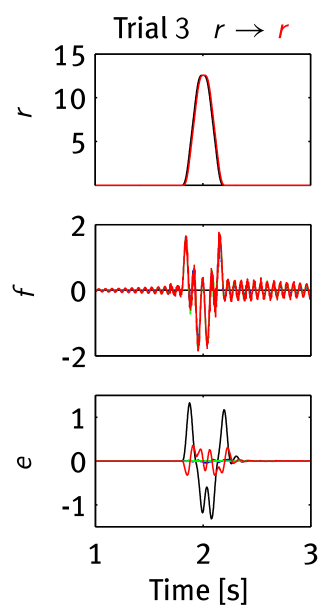
- Trial-invariant task
- Identical Initial conditions

Example experiment

- Perform trials
- Change task $r \rightarrow r$

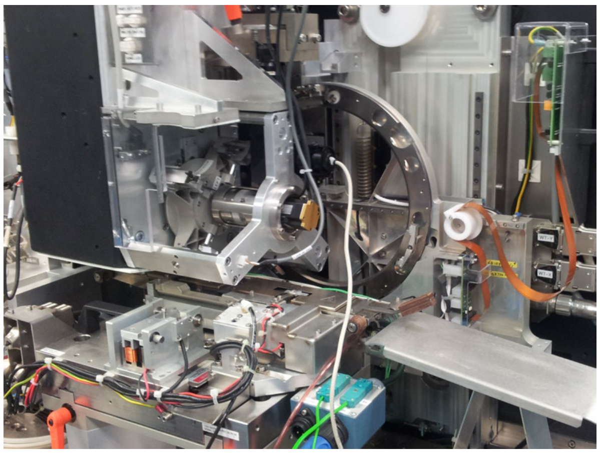
Consequences

- Control signal optimal for *specific* task only
- Performance deteriorates for varying tasks



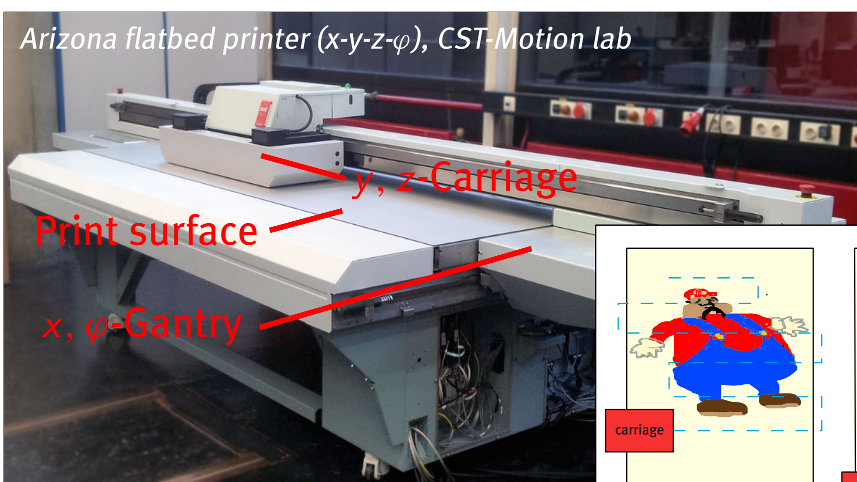
Application examples with reference variations

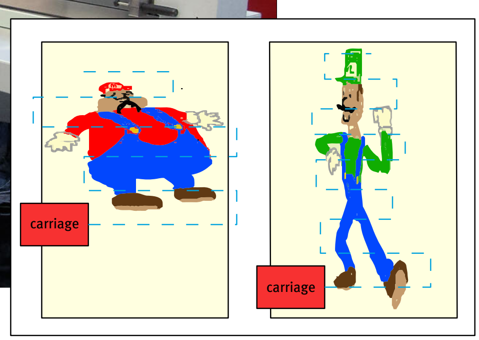
- Diebonding at NXP ITEC (Boeren et al. 2016)



Application examples with reference variations

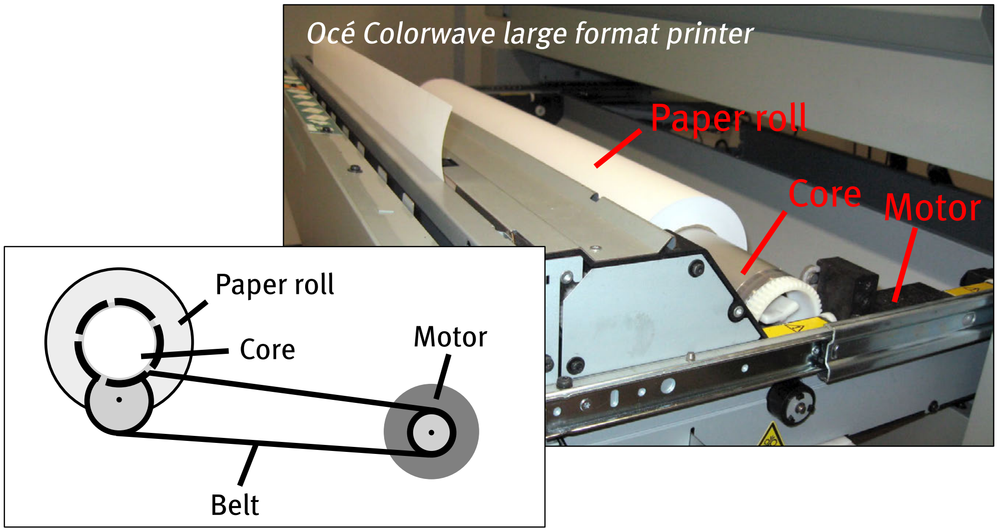
- Industrial printing





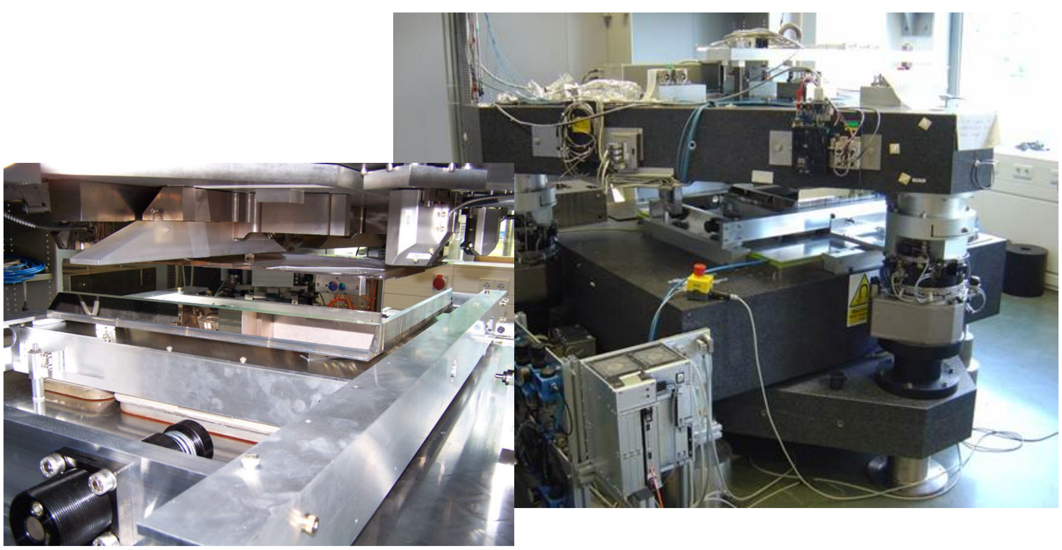
Application examples with reference variations

- Industrial printing



Application examples with reference variations

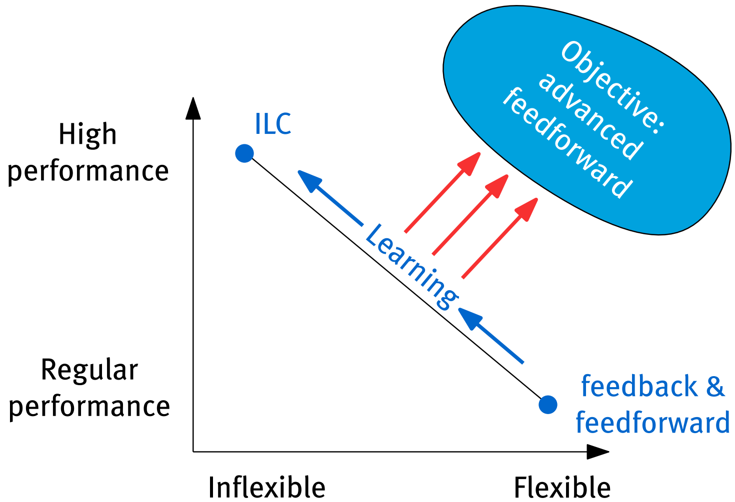
- Wafer positioning at Philips (Blanken et al. 2017)



Introduction

Requirements in many motion systems

- Performance: accuracy and speed
- Flexibility: variation in tasks



Introduction

Towards learning feedforward

- Performance improvements w.r.t. manual tuning
- Adapt to system variations, e.g., motor temperature
- Per system tuning, e.g., for mass-produced printers
- Enable complex ff structures, e.g., MIMO, rational
- Many recent developments, see e.g., van der Meulen et al. (2008), van de Wijdeven & Bosgra (2010), Boeren et al. (2014, 2015), Blanken et al. (2017), Blanken & Oomen (2020)

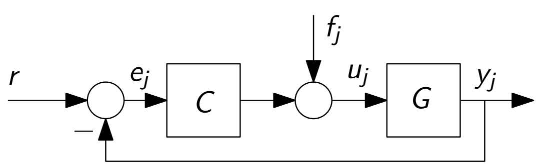
This lecture

1. Feedforward parametrization
2. Automated tuning with optimal ILC
3. Convergence and robustness
4. Design example and extensions
5. Exercise with experiments

Feedforward parameterization

Control setup

- Trial index $j = 0, 1, \ldots$
- Signal vectors $r, e_j, f_j, u_j, y_j \in \mathbb{R}^n$



Lifted notation

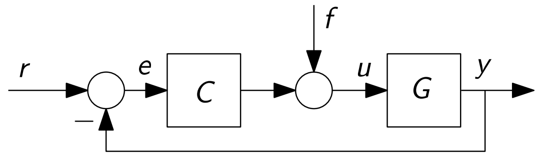
$$\underbrace{\begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N-1] \end{bmatrix}}_{y} = \underbrace{\begin{bmatrix} h(0) & 0 & \ldots & 0 \\ h(1) & h(0) & \ldots & 0 \\ \vdots & & \ddots & \vdots \\ h(N-1) & h(n-2) & \ldots & h(0) \end{bmatrix}}_{G} \underbrace{\begin{bmatrix} u[0] \\ u[1] \\ \vdots \\ u[N-1] \end{bmatrix}}_{u}$$

- Impulse response $h(k)$
- $y_j = Gu_j$

Feedforward parameterization

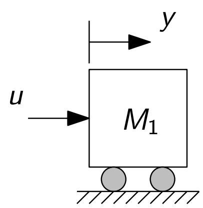
Feedforward example

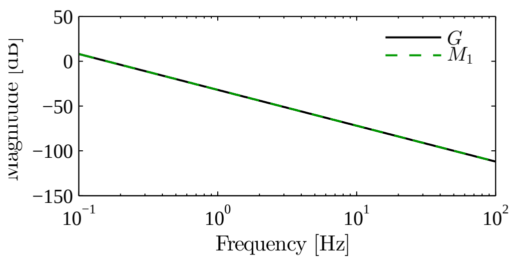
- Q: how to choose f?
- A: recall $G = \frac{1}{M_1 s^2}$, hence



$$e = S(r - Gf)$$
$$f = G^{-1}r \to e = 0$$

- Implement acceleration feedforward $f = M_1 s^2 r$

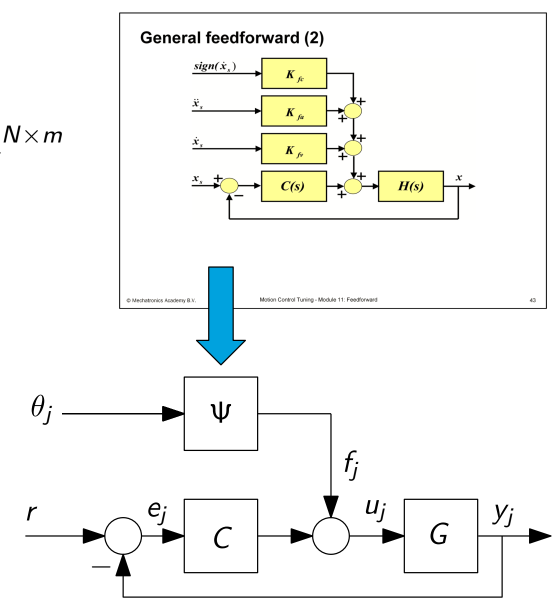




Feedforward parameterization

Feedforward parameterization

- Cast typical feedforward choices in general structure
- $f_j = \Psi\theta_j$
- Basis $\Psi = [\varphi_1, \ \varphi_2, ..., \varphi_m] \in \mathbb{R}^{N \times m}$
- Parameters θ_j
- Basis can be reference dependent $\Psi(r)$

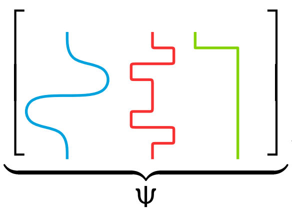


Feedforward parameterization

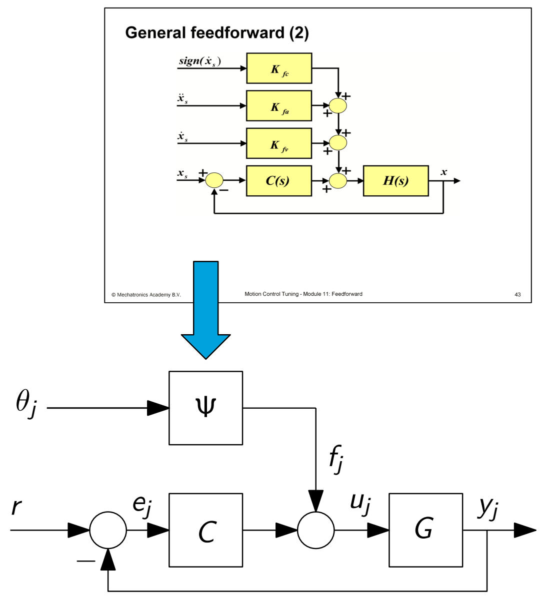
Example feedforward parameterization

- Velocity, acceleration, and friction compensation
- In basis function notation $f_j = \Psi\theta_j$ with

$$\underbrace{\begin{bmatrix} f_j[0] \\ f_j[1] \\ \vdots \\ f_j[N-1] \end{bmatrix}}_{f_j} = \underbrace{\Big[\quad\Big]}_{\Psi} \underbrace{\begin{bmatrix} \theta_j[0] \\ \theta_j[1] \\ \theta_j[2] \end{bmatrix}}_{\theta_j}$$



- Resulting feedforward
  $f_j = \dot{r} \cdot \theta_j[0] + \ddot{r} \cdot \theta_j[1] + \text{sign}(\dot{r}) \cdot \theta_j[2]$
- Note f_j linear in θ_j and *nonlinear* in r



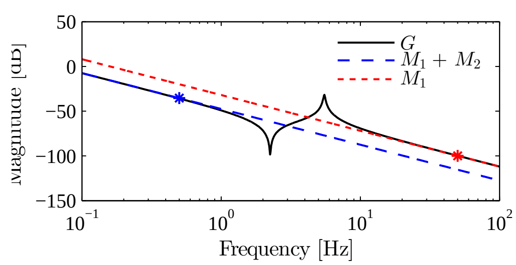
Feedforward parameterization

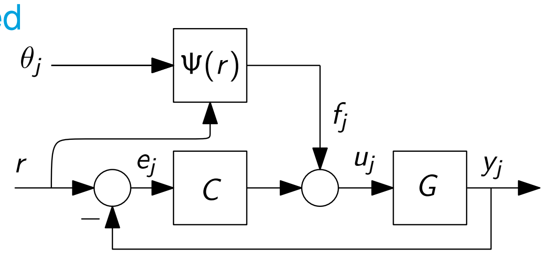
Feedforward example - revisited

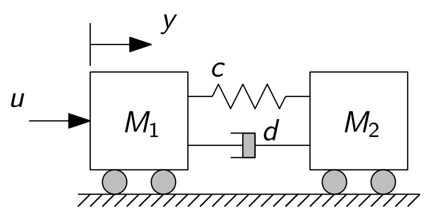
- Suppose acceleration ff
  $\Psi(r) = \ddot{r}$ hence $f_j = \ddot{r}\theta_j$
- Q: best value for θ_j?
- A: depends on reference

$r^a = sin(0.5 \cdot 2\pi t)$

$r^b = sin(50 \cdot 2\pi t)$



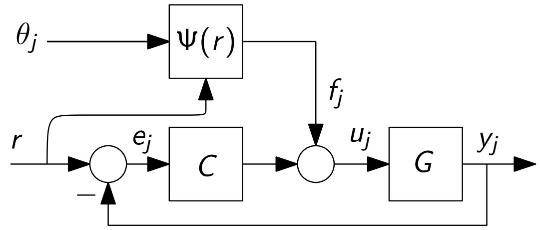




Feedforward parameterization

Feedforward example - revisited

- $e_j = S(r - G\Psi\theta_j)$
- If $\Psi\theta_j \neq G^{-1}r \to e \neq 0$
- Undermodeling in $\Psi \to$ bias



Consequences in case of significant bias

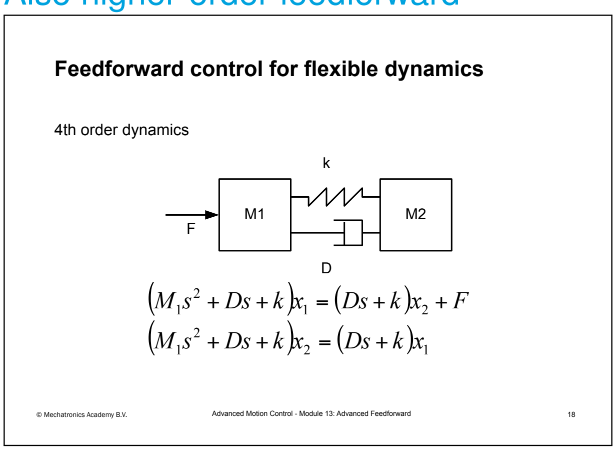
- Optimal θ_j depends on reference $\Rightarrow$ poor extrapolation capabilities
- Also in classical feedforward design(Oomen 2020)

Feedforward parameterization design goal

1. Design $\Psi(r)\theta_j \approx G^{-1}r$ to eliminate bias and achieve performance for varying references
2. In case of nonzero e_j, shape e_j using weighting matrices in norm-optimal ILC
   - e.g., emphasize constant velocity part (printing, wafer scanning)(Oomen & Rojas 2017)

Feedforward parameterization

Also higher-order feedforward



Feedforward parameterization

Also higher-order feedforward

Feedforward control for flexible dynamics

4th order system (measured on load)

$$G = \frac{(Ds+k)}{M_1 M_2 s^2\left(s^2 + \frac{M_1+M_2}{M_1 M_2} Ds + \frac{M_1+M_2}{M_1 M_2} k\right)}$$
$$= \frac{(Ds+k)}{M_1 M_2 s^2\left(s^2 + 2\xi\varpi s + \varpi^2\right)}$$
$$\Rightarrow K_{FF} = G^{-1} = \frac{M_1 M_2 s^2\left(s^2 + 2\xi\varpi s + \varpi^2\right)}{(Ds+k)}$$

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Feedforward parameterization

Also higher-order feedforward

Feedforward control for flexible dynamics

Low damping :

$$\Rightarrow K_{FF} = \frac{M_1 M_2 s^2\left(s^2 + \varpi^2\right)}{k} = \frac{M_1 M_2}{k} s^4 + \frac{M_1 M_2 \varpi^2}{k} s^2$$
$$= \underbrace{\frac{M_1 M_2}{k} s^4}_{\text{snap FF}} + \underbrace{(M_1 + M_2) s^2}_{\text{acceleration FF}}$$

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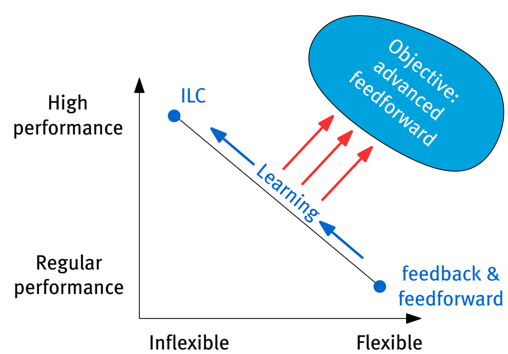


- many parameters (masses)
- higher-order derivatives can be directly used to learn the coefficients!

Tuning feedforward parameters with ILC

Optimal ILC synthesis

- Performance criterion:

$$\mathcal{J}(\theta_{j+1}) := ||e_{j+1}||^2_{W_e} + ||f_{j+1}||^2_{W_f} + ||f_{j+1} - f_j||^2_{W_{\Delta f}}$$

with W_e, W_f, $W_{\Delta f}$ positive (semi-)definite and $||x||_W = \sqrt{x^T W x}$.

Learning update with basis functions

- Analytic solution by solving $\frac{\partial \mathcal{J}(\theta_{j+1})}{\partial \theta_{j+1}} = 0$
- $\theta_{j+1} = Q\theta_j + Le_j$ with

$$L = \left(\Psi^T \left(J^T W_e J + W_f + W_{\Delta f}\right) \Psi\right)^{-1} \Psi^T J^T W_e$$

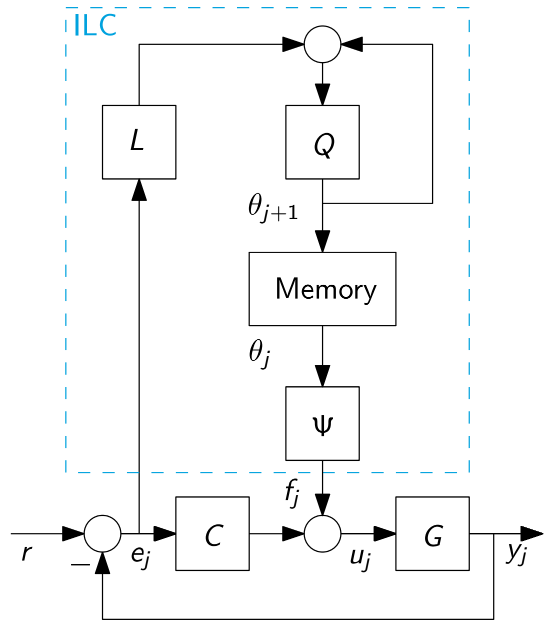
$$Q = \left(\Psi^T \left(J^T W_e J + W_f + W_{\Delta f}\right) \Psi\right)^{-1} \Psi^T \left(J^T W_e J + W_{\Delta f}\right) \Psi$$

- Set $\Psi = I$ to recover standard norm-optimal ILC

Tuning feedforward parameters with ILC

Implementation ILC with basis functions

- extension of standard ILC
- Typically less memory required
- Faster computations L and Q: $\mathcal{O}(m^3N)$ vs. $\mathcal{O}(N^3)$, $m << N$
- Enable reference variations, recompute L and Q if $\Psi(r)$ changes
- Highly robust convergence (next)



Tuning feedforward parameters with ILC

Improving convergence and robustness w.r.t. standard ILC

- Monotonic convergence:

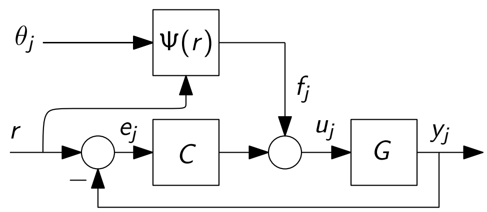
$$\bar{\sigma}(Q - LJ\Psi) = \bar{\sigma}\left(\left(\Psi^T\left(J^TW_eJ + W_f + W_{\Delta f}\right)\Psi\right)^{-1}\Psi^TW_{\Delta f}\Psi\right) < 1$$

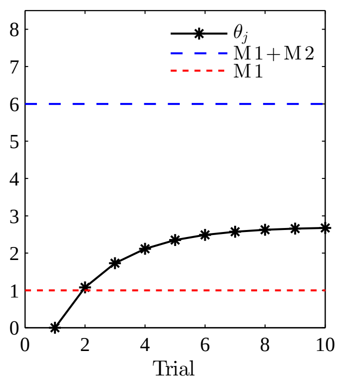
- Results
  1. If $\Psi^T\left(J^TW_eJ + W_f + W_{\Delta f}\right)\Psi \succ 0$, guaranteed monotonic convergence
  2. If J singular then exploit freedom in Ψ or set $W_f \succ 0$ to guarantee monotonic convergence
  3. Basis functions increase robustness, see (Boeren et al. 2016)

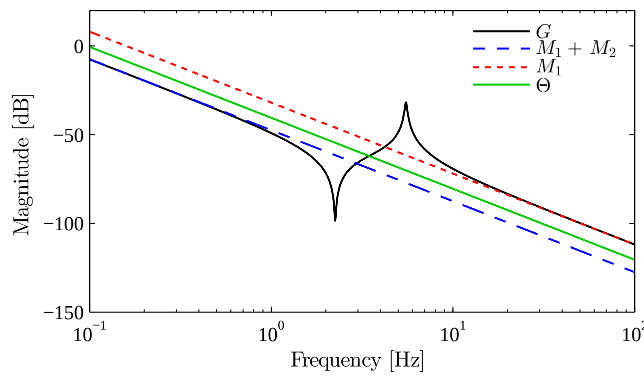
Tuning feedforward parameters with ILC

Feedforward example - revisited

- $\Psi = \ddot{r}$ hence $f_j = \ddot{r}\theta_j$
- $W_e = I$, $W_f = W_{\Delta f} = 0$
- Simulation with $\theta_{j+1} = Q\theta_j + \alpha L e_j$ and $\alpha = 0.5$







Feedforward parameterization

Tuning feedforward parameters with ILC

Design example with varying references

Towards rational parameterizations and input shaping

Application to Arizona

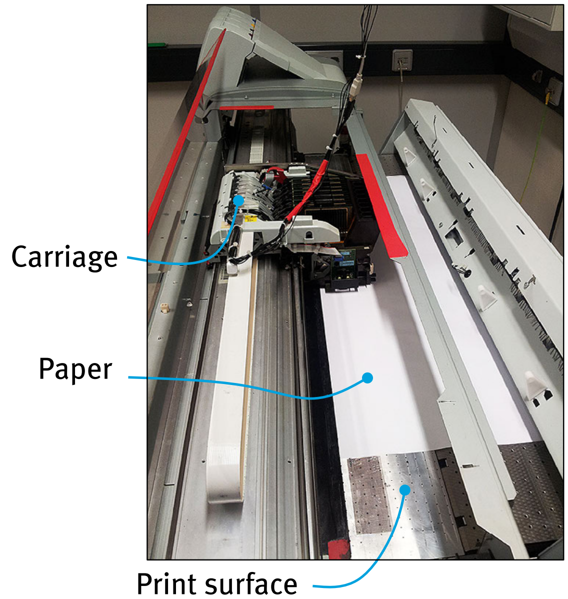
Conclusions

Extensions

Design example with varying references

Wide-format printing

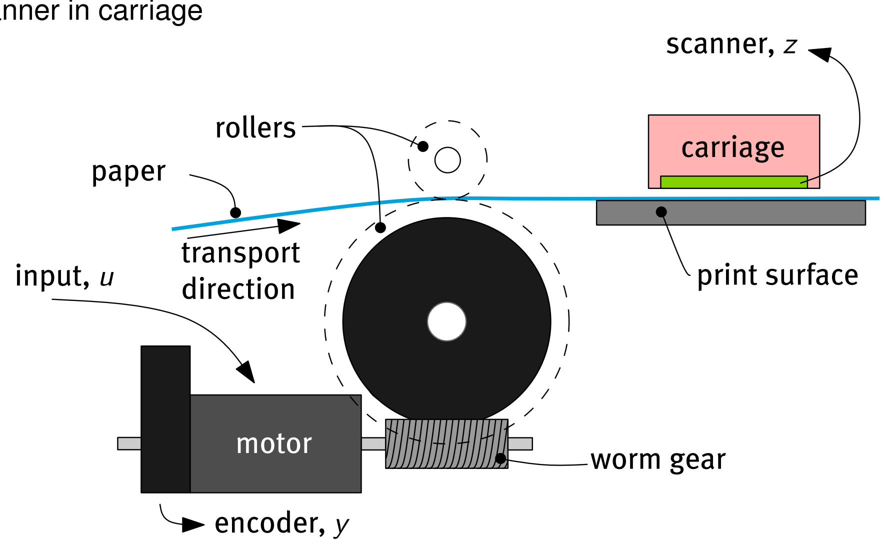
- Control of paper position
- Objective:
  achieve accurate positioning
- See (Bolder et al. 2014)
  for present study
- Outline
  1. System and control setup
  2. Design basis
     Tune weighting matrices
  3. Experimental results



Design example with varying references

Medium positioning drive

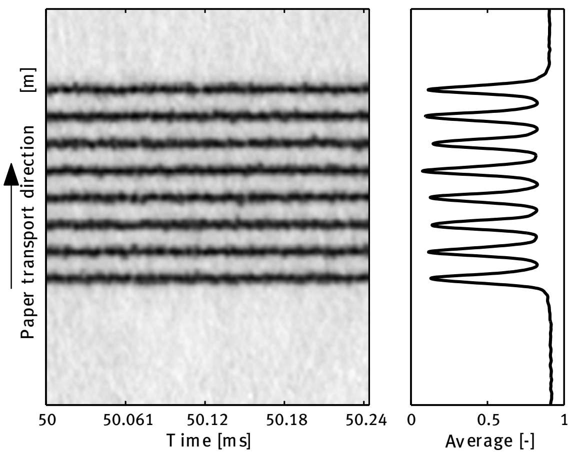
- Motor with encoder
- Scanner in carriage



Design example with varying references

Segment of a scan

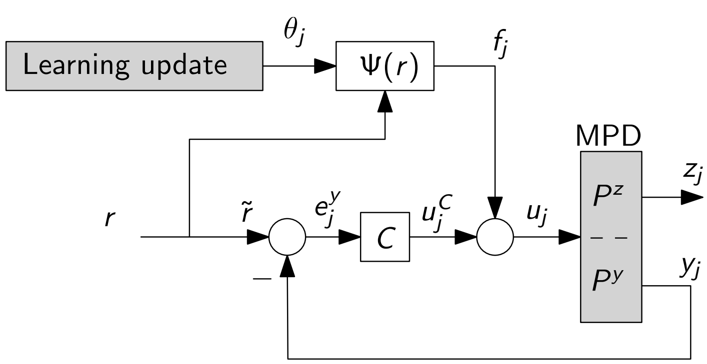
- Accurately measure paper position
- Offline only: not suited for real-time feedback control
- Enables direct performance measurement in ILC



Design example with varying references

Control setup

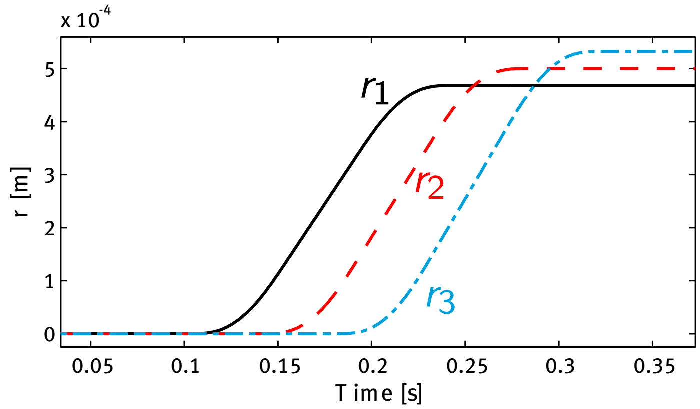
- Feedback uses encoder measurements
- ILC using scanner $e_j^z = r - z_j$
- Basis $\Psi(r)$ for varying r



Design example with varying references

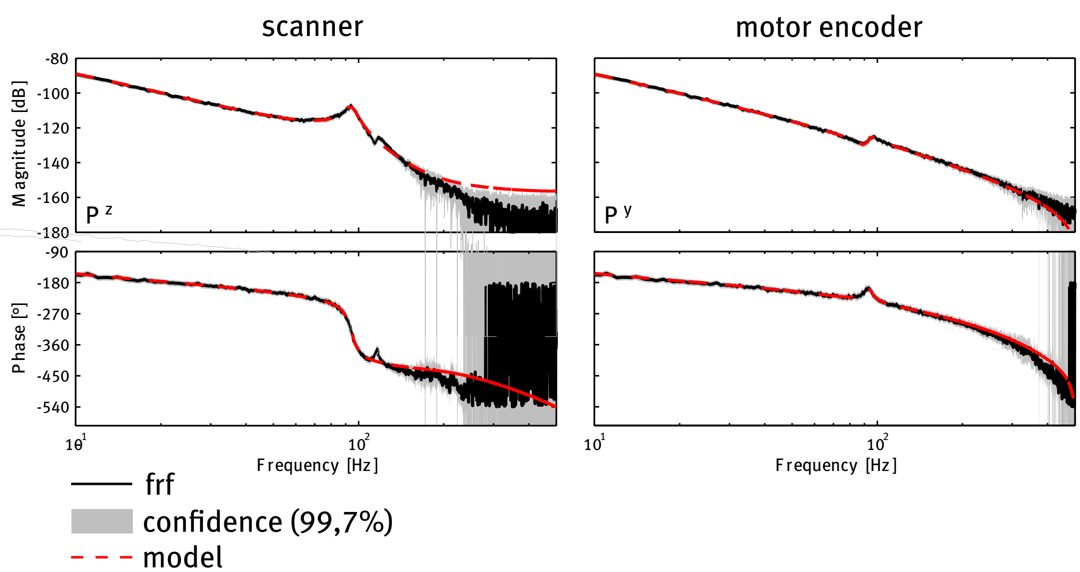
Benchmark references

- Demonstrate required accuracy
- Comparison ILC and ILC with basis functions



Design example with varying references

Frequency response and model



Design example with varying references

Basis design

- Design goal $\Psi(r)\theta_j \approx G^{-1} r$
- Analysis of scanner FRF:
  1. Two stable real-valued poles below 40 Hz $\rightarrow$ rigid-body dynamics, viscous friction, and counter-electromotive force
  2. Two stable complex-conjugated poles at 94 Hz $\rightarrow$ resonance roller and DC-motor.
  3. Two minimum-phase complex-conjugated zeros at 115 Hz $\rightarrow$ anti-resonance roller and DC-motor. (ignored)
  4. Two stable complex-conjugated poles at 119 Hz $\rightarrow$ flexible connection roller encoder (ignored)
- $\Psi(r) = \left[r, \dot{r}, \ddot{r}, \frac{d^3}{dt^3} r, \frac{d^4}{dt^4} r \right]$
- Compensation of position, velocity, acceleration, jerk, and snap effects, see (Lambrechts et al. 2005)

Design example with varying references

Weighting matrix tuning

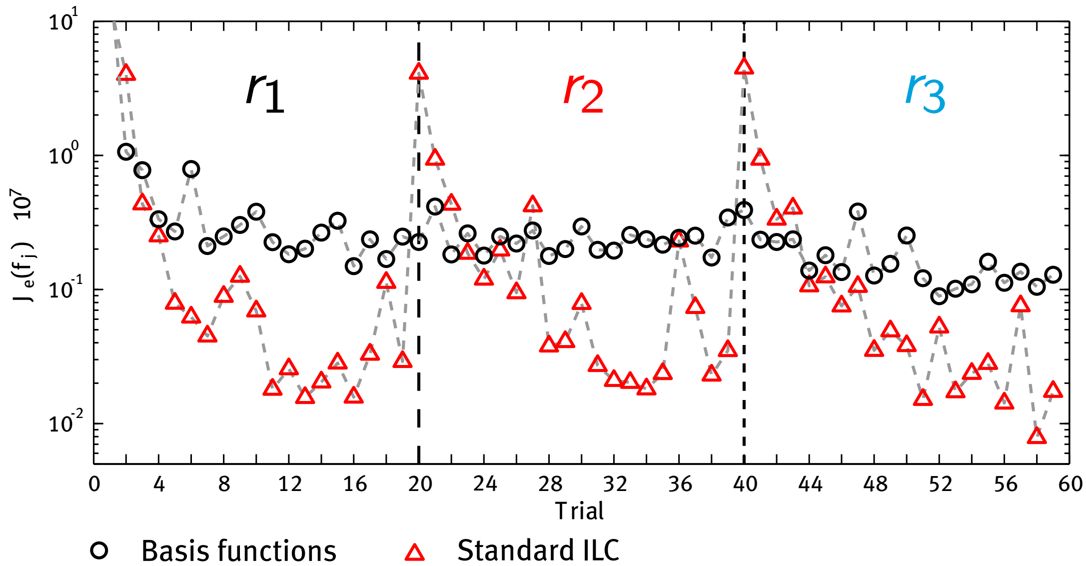
- Tuning to achieve smallest e_j
- Used constant-diagonal matrices for simplicity
- Robustness
  1. $W_e = I \cdot 10^6$, $W_{\Delta f} = 0$, and $W_f = Ic_1$ with $c_1 >> 1$
  2. Decreased c_1
  3. Model accurate enough to set $W_f = 0$
- Trial-varying disturbances vs. learning speed
  1. $W_{\Delta f} = Ic_2$, with $c_2 = 0$
  2. Gradually increase c_2
- Result

$$W_e = I \cdot 10^6, \; W_f = 0, \text{ and } \; W_{\Delta f} = I \cdot 10^{-3}.$$

- High robustness of basis function approach often allows straightforward minimal tracking error choice of $W_e = I$, $W_f = 0$, $W_{\Delta f} = 0$

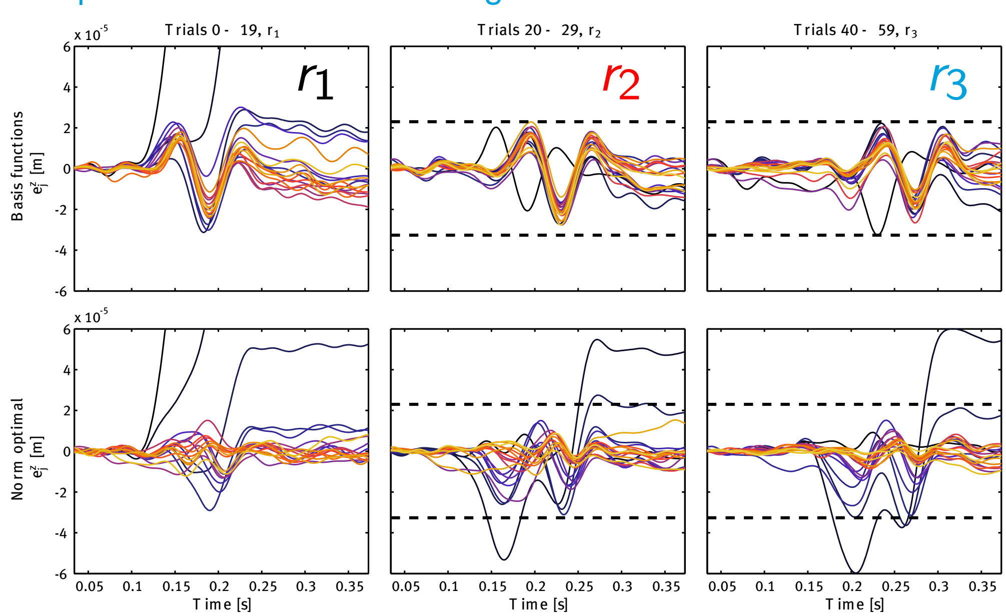
Design example with varying references

Experimental results - cost function



Design example with varying references

Experimental results - tracking errors



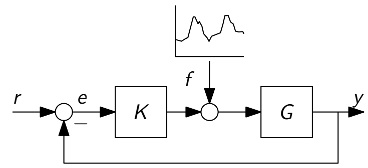
Concluding design example

- Demonstrated consistent tracking performance in 20 μm band
- Main trail-varying disturbances: outliers image processing
- Main remaining uncompensated effects: cyclic roller effects, medium friction (stick)
- Further accuracy improvements by extending $\Psi(r)$ possible
- Further weighting matrix tuning enables *shaping* e_j

Towards rational parameterizations and input shaping

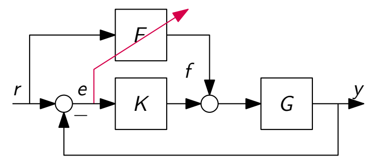
Flexibility in ILC

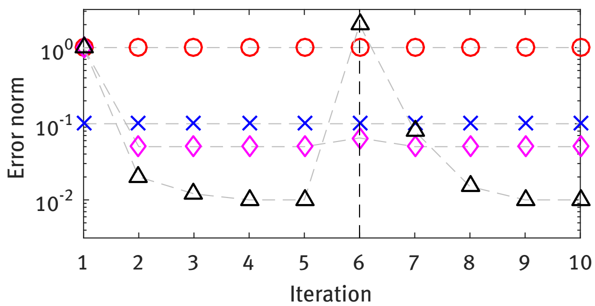
- Traditional ILC is inflexible to varying references...
  ...since it iteratively learns a signal f

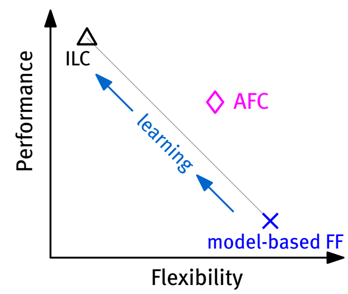


Advanced feedforward control

1. Iteratively learn controller $F(\theta) = \theta\Psi$,
   e.g., $\Psi = \frac{d^2}{dt^2}$ (mass feedforward)





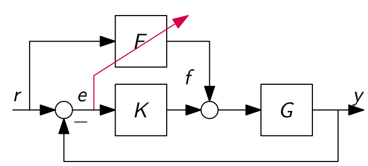


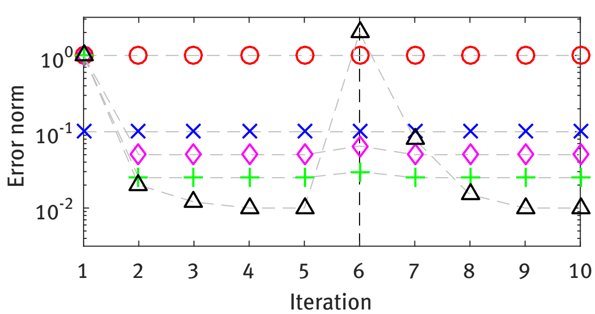
Towards rational parameterizations and input shaping

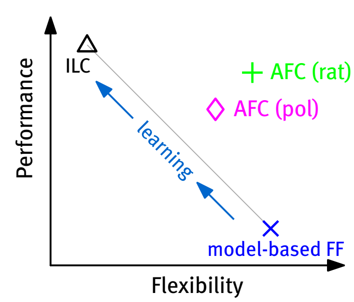
Advanced feedforward control

Iteratively learn controller F instead of signal f

2. Polynomial: $F = \theta\Psi$
3. Rational: $F = \frac{\theta^A \Psi^A}{\theta^B \Psi^B}$
   - Nonlinear in parameters
   - Solutions: ILC [Bolder, PhD 2015], IV-tuning [Boeren, PhD 2016]
   - Key aspect: enables pre-actuation



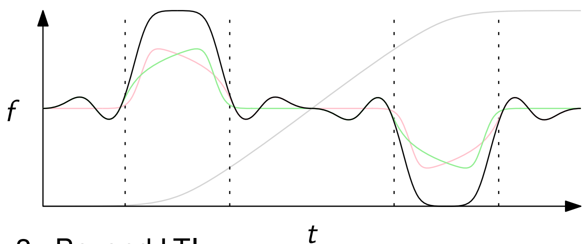


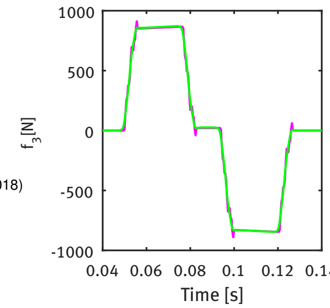


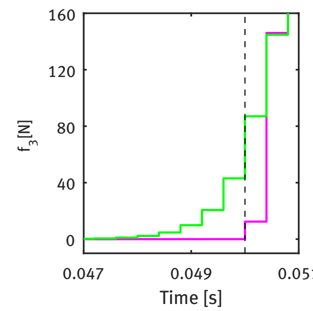
Towards rational parameterizations and input shaping

Pre-actuation in advanced feedforward control

2. Rational: $F = \frac{\theta^A \Psi^A}{\theta^B \Psi^B}$
   - Unstable poles? $\rightarrow$ non-causal!
     Use the bilateral Laplace/Z-transform (instead of unilateral)
     Exact solution (‘ZPETC’ is approximate)
   - Non-causal feedforward enables pre-actuation (Blanken et al. 2017) (van Zundert & Oomen 2018)





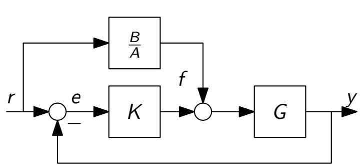


3. Beyond LTI:
   - approaches: LTV (van Zundert et al. 2016), LPV (de Rozario et al. 2017)
   - applications: position-dependent systems, flexible sampling, nonlinear, ...

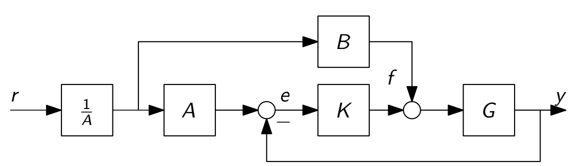
Towards rational parameterizations and input shaping

The algorithm

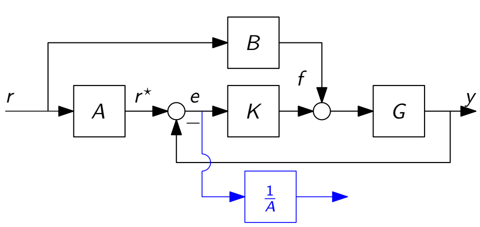
1. Original



2. Rewrite



3. Commute



- rational ff (Bolder & Oomen 2015)
  - solve black in Step 3 (linear in parameters)
  - iterate over blue in Step 3
  - implement as in Step 1
- input shaping (Boeren et al. 2014)
  - ignore the blue part
  - $r = r^*$ after n_A samples (order FIR filter A)
  - point-to-point motion tasks!
  - linear in parameters, no stability issues
    - e.g., sampling zero, take $A = (1 - z^{-1})$
    - e.g., delay compensation: $A = z^{-n_d}$

Feedforward parameterization

Tuning feedforward parameters with ILC

Design example with varying references

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Application to Arizona

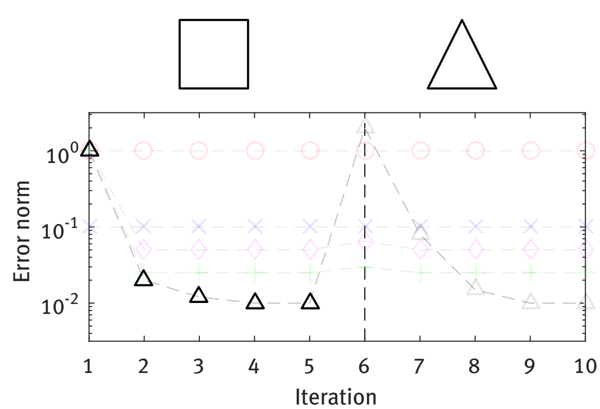
Conclusions

Extensions

Application to Arizona

Advanced feedforward control - drawing conclusions

What do these norm plots mean? Let's draw some conclusions on our printer!



- ILC: high performance through learning, but inflexible!
- AFC: high flexibility, while maintaining good performance
- can we further improve performance of AFC?

Conclusions

Conclusions

- basis functions allow combine data-driven learning and model-based feedforward:
  - high performance and task flexibility
- convenient for tuning in industrial implementations ('automated parameter calibration')
- many ongoing developments
  - general structures (Blanken et al. 2017)
  - automated structure selection (Oomen & Rojas 2017)
  - Gaussian process parameterizations (Blanken & Oomen 2020)
  - MIMO (next slides)
  - ...
  - ...

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Extensions

ILC with basis functions for multivariable systems

- Developed framework directly applicable to MIMO situation
- See (van Deursen 2015, Eijsermans 2014) for results

Basis design

$$\Psi = \begin{bmatrix} \ddot{r}_x(r_y) & \dot{r}_x(r_y) & r_x(r_y) & \dddot{r}_x(r_y) & f_{PS}(r_y) & f_{PL}(r_y) & r_x^{\frac{1}{2}} & r_x^{\frac{1}{4}} & r_x^{\frac{1}{6}} & & & & & & & & & & & & & \\ & & & & & & & & & \ddot{r}_x(r_y) & \dot{r}_x(r_y) & r_x(r_y) & \dddot{r}_x(r_y) & \ddot{r}_y & f_{PS}(r_y) & f_{PL}(r_y) & r_x^{\frac{1}{4}} & r_x^{\frac{1}{2}} & r_x & r_x^2 & & \\ & & & & & & & & & & & & & & & & & & & & \dot{r}_y & \ddot{r}_y \end{bmatrix}$$

- First principles, combined with data
- Cable chain compensation
- Remaining unknown reproducible effects

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